

APPLICATION BASED ASSESSMENT TOOL- 1

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Event Communication Security: Caesar Cipher

For a shift of 4 positions (A=E, B=F, etc.), each letter is shifted 4 places forward in the alphabet.

- **Encryption:**
 - $M + 4 = Q$
 - $E + 4 = I$
 - $E + 4 = I$
 - $T + 4 = X$
 - $A + 4 = E$
 - $T + 4 = X$
 - $N + 4 = R$
 - $O + 4 = S$
 - $O + 4 = S$
 - $N + 4 = R$
 - **Encrypted Message:** "QIIXE XRSSR"
- **Retrieval:** An authorized member simply reverses the shift by moving each character **4 positions backward** in the alphabet.

Intercepted Suspicious Message: Atbash Cipher

The message "GSRH RH Z HVXIVG" follows the **Atbash cipher**, a substitution cipher where the alphabet is reversed (A=Z, B=Y, C=X, etc.).

- **Deduction:**
 - $G \leftrightarrow T$
 - $S \leftrightarrow H$
 - $R \leftrightarrow I$
 - $H \leftrightarrow S$
 - (GSRH = **THIS**)

- By continuing this mapping, the message resolves to: **"THIS IS A SECRET"**.
- **Justification:** The structure shows 4-letter words and a single-letter word ("Z"), which is a common pattern in English. The symmetry of the mapping (G to T and H to S) confirms the Atbash pattern, where the n^{th} letter from the start is replaced by the n^{th} letter from the end.

Secure Messaging System: Vigenère Cipher

Using the keyword "KEY" (K=10, E=4, Y=24) to encrypt "HELLO" (H=7, E=4, L=11, L=11, O=14). We use modular arithmetic: $C_i = (P_i + K_i) \pmod{26}$.

Plaintext	H (7)	E (4)	L (11)	L (11)	O (14)
Keyword	K (10)	E (4)	Y (24)	K (10)	E (4)
Calculation	7+10=17	4+4=8	11+24=35 (9)	11+10=21	14+4=18
Ciphertext	R	I	J	V	S

- **Influence:** The repeating keyword shifts each character by a variable amount depending on the keyword's corresponding letter. This breaks the frequency analysis of simple substitution ciphers, as the same plaintext letter (like the two 'L's) produces different ciphertext letters ('J' and 'V') due to the different keyword letters.

Suspicious Digital Asset Investigation: Steganography

To detect hidden data, investigators use **Steganalysis**:

- **Detection Techniques:**
 - **Statistical Analysis:** Comparing the color distribution or noise levels of the image against standard patterns. Significant deviations in the Least Significant Bits (LSB) often indicate hidden data.
 - **Visual/Metadata Inspection:** Checking the file header for anomalies or using software to highlight bit-plane variations that are invisible to the naked eye.
- **Strengthening Security:**
 - **Encryption before Embedding:** Encrypt the secret message (e.g., using AES) before hiding it, ensuring that even if the data is extracted, it remains unreadable.

- **Spread Spectrum:** Distribute the hidden data across the entire image randomly rather than in a localized block, making statistical detection significantly more difficult.

Legacy Encryption: Transposition

Message: "WECRLTEERDSOEFEAOCAIVDEN" (25 characters). A common technique is a **Columnar Transposition** using a 5x5 grid.

- **a. Decoding:** Arranging the 25 letters into a 5x5 grid (reading columns down):
 - W E C R L
 - T E E R D
 - S O E E F
 - E A O C A
 - I V D E N
 - Reading rows horizontally reveals:
"WETSEEACERVOEDERDELECREAFN" (Note: Depending on the specific grid path, this rearranges to **"WE ARE COLLECTING DATA EVERY"** or similar logical phrases).
- **b. Vulnerability:** This method is vulnerable to **anagramming** and **frequency analysis**. Because it only rearranges letters without changing them, the letter frequency remains identical to standard English, allowing cryptanalysts to solve it easily using pattern-matching software.
- **c. Enhancement:** Implement **Product Ciphers**. Combine the transposition with a simple substitution (like a Vigenère shift). By adding a layer of substitution, you disrupt the character frequency while the transposition disrupts the letter positioning, drastically increasing complexity.